SUPERMARKET POS SYSTEM

BY PRIME LOGIC

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SUPERMARKET POS SYSTEM

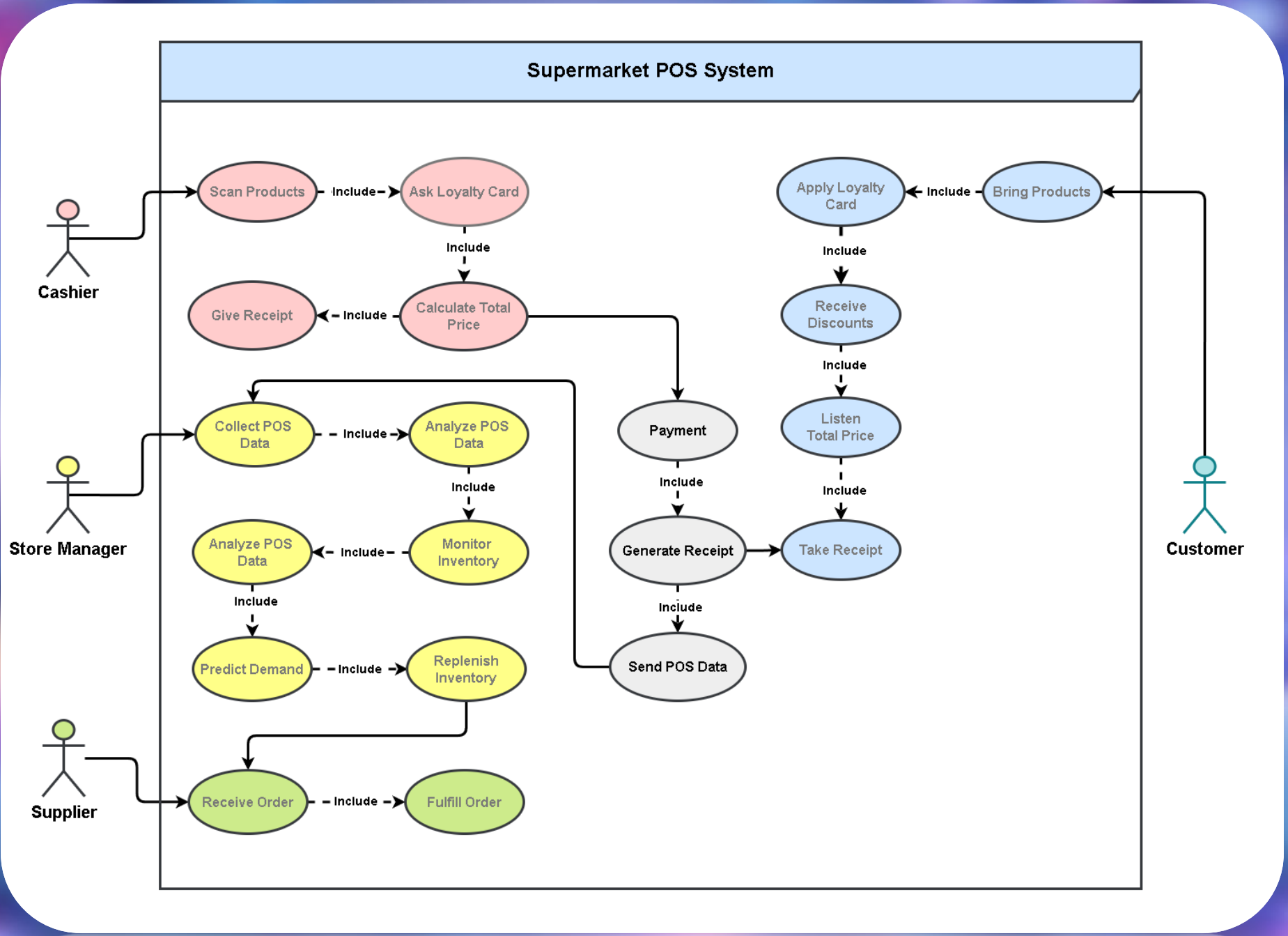
Is an Information Systems that revolves around the data provided by POS machines, and is used to monitor, analyze supermarket inventory and predict demand for individual products by the Store Manager.

USE CASE

Overall, there are 4 actors:

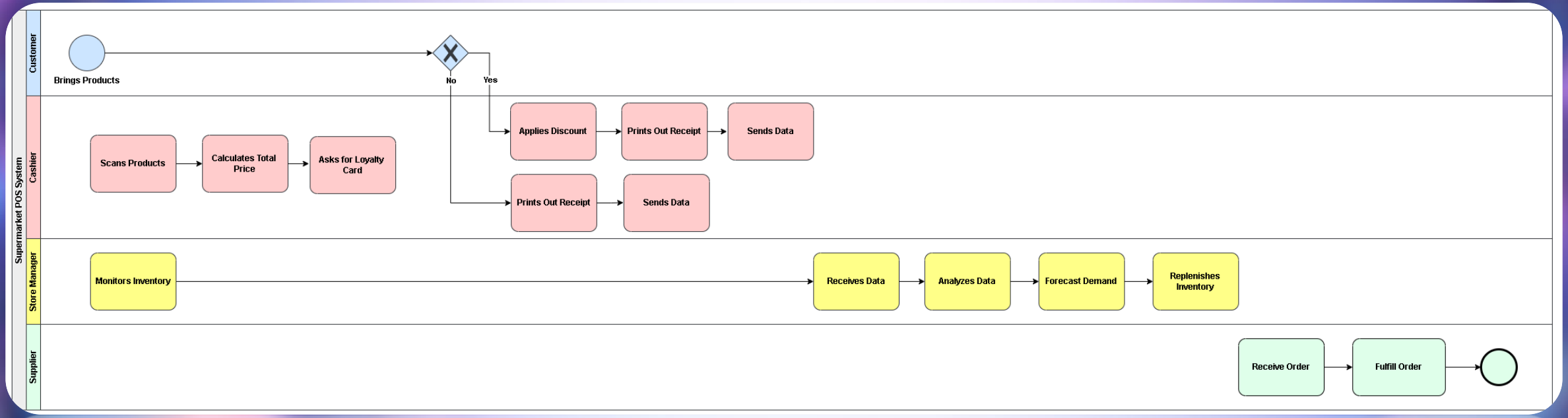
1. Customer,
2. Cashier,
3. Store Manager with
4. Supplier

performing 20 distinct operations.


Supermarket POS System
Cashier
Store Manager
Supplier
Customer
Scan Products
Include
Ask Loyalty Card
Include
Give Receipt
Include
Calculate Total Price
Collect POS Data
Include
Analyze POS Data
Include
Analyze POS Data
Include
Monitor Inventory
Include
Predict Demand
Include
Replenish Inventory
Receive Order
Include
Fulfill Order
Payment
Include
Generate Receipt
Include
Send POS Data
Apply Loyalty Card
Include
Bring Products
Include
Receive Discounts
Include
Listen Total Price
Include
Take Receipt

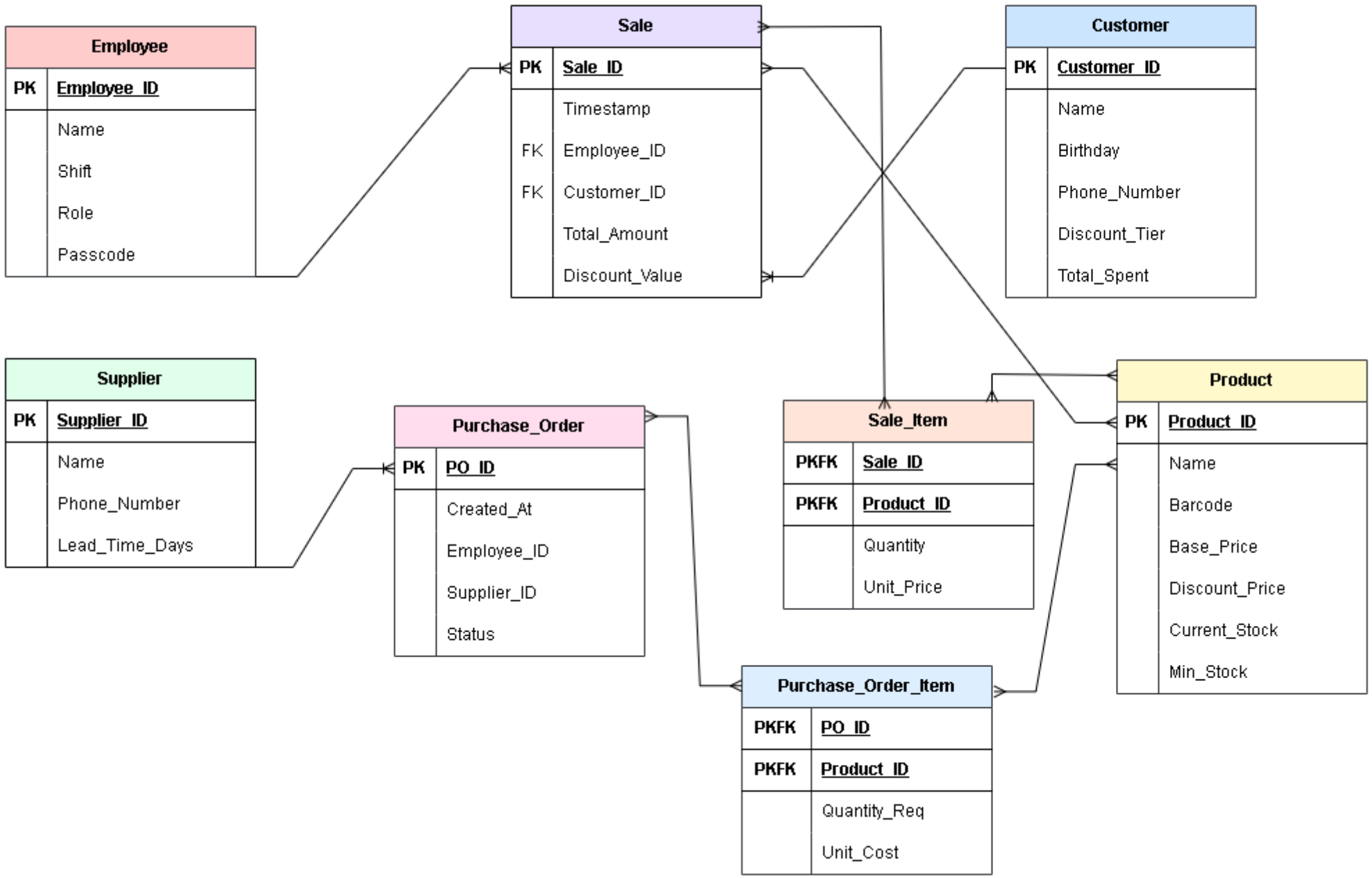
BPM

There are 18 operations and 1 conditional. Customer notably performs the smallest number of operations (1), followed closely by supplier (2). On the contrary, Cashier stands out with the highest number of operations accounting for 8.


Supermarket POS System
Customer
Cashier
Store Manager
Supplier
Brings Products
No
Yes
Scans Products
Calculates Total Price
Asks for Loyalty Card
Applies Discount
Prints Out Receipt
Sends Data
Prints Out Receipt
Sends Data
Monitors Inventory
Receives Data
Analyzes Data
Forecast Demand
Replenishes Inventory
Receive Order
Fulfill Order

ERD

There are total 8 entities: 4 independent (Customer, Employee, Sale, Product), 2 dependent (Sale, Purchase Order), and 2 associative (Sale Item, Purchase Order Item).


Employee
PK Employee_ID
Name
Shift
Role
Passcode
Sale
PK Sale_ID
Timestamp
FK Employee_ID
FK Customer_ID
Total_Amount
Discount_Value
Customer
PK Customer_ID
Name
Birthday
Phone_Number
Discount_Tier
Total_Spent
Supplier
PK Supplier_ID
Name
Phone_Number
Lead_Time_Days
Purchase_Order
PK PO_ID
Created_At
Employee_ID
Supplier_ID
Status
Sale_Item
PKFK Sale_ID
PKFK Product_ID
Quantity
Unit_Price
Product
PK Product_ID
Name
Barcode
Base_Price
Discount_Price
Current_Stock
Min_Stock
Purchase_Order_Item
PKFK PO_ID
PKFK Product_ID
Quantity_Req
Unit_Cost

SQL CODING

We've conducted vibe coding using an AI to create SQL database, and the mentioned 8 entities: 4 independent, 2 dependent, and 2 associative.

```
mysql> show databases;
+--------------------+
| Database           |
+--------------------+
| information_schema |
| mysql              |
| performance_schema |
| prime_logic        |
| sys                |
+--------------------+
5 rows in set (0.00 sec)
```

CUSTOMER

Customer table has 4 rows: phone number, discount tier, and total money spent accompanied by customer ID being the primary key.

```
mysql> DESCRIBE customer;
+----------------+---------------+------+-----+---------+----------------+
| Field          | Type          | Null | Key | Default | Extra          |
+----------------+---------------+------+-----+---------+----------------+
| Customer_ID    | int           | NO   | PRI | NULL    | auto_increment |
| Phone_Number   | varchar(15)   | YES  | UNI | NULL    |                |
| Discount_Tier  | decimal(3,2)  | NO   |     | NULL    |                |
| Total_Spent    | decimal(10,2) | YES  |     | 0.00    |                |
+----------------+---------------+------+-----+---------+----------------+
4 rows in set (0.01 sec)
```

EMPLOYEE

Employee table has 5 rows: first name, last name, and role accompanied by passcode hash with employee ID being the primary key.

```
mysql> DESCRIBE Employee;
+---------------+-----------------------------------+------+-----+---------+----------------+
| Field         | Type                              | Null | Key | Default | Extra          |
+---------------+-----------------------------------+------+-----+---------+----------------+
| Employee_ID   | int                               | NO   | PRI | NULL    | auto_increment |
| First_Name    | varchar(50)                       | NO   |     | NULL    |                |
| Last_Name     | varchar(50)                       | NO   |     | NULL    |                |
| Role          | enum('Cashier','Manager','Admin') | NO   |     | NULL    |                |
| Passcode_Hash | varchar(255)                      | NO   |     | NULL    |                |
+---------------+-----------------------------------+------+-----+---------+----------------+
5 rows in set (0.00 sec)
```

PRODUCT

Product table has 6 rows: barcode, name, and price accompanied by available stock with minimal stockhold followed by product ID being the primary key.

```
mysql> describe product;
+---------------+---------------+------+-----+---------+----------------+
| Field         | Type          | Null | Key | Default | Extra          |
+---------------+---------------+------+-----+---------+----------------+
| Product_ID    | int           | NO   | PRI | NULL    | auto_increment |
| Barcode       | varchar(50)   | NO   | UNI | NULL    |                |
| Product_Name  | varchar(100)  | NO   |     | NULL    |                |
| Base_Price    | decimal(10,2) | NO   |     | NULL    |                |
| Current_Stock | int           | NO   |     | NULL    |                |
| Min_Threshold | int           | NO   |     | NULL    |                |
+---------------+---------------+------+-----+---------+----------------+
6 rows in set (0.00 sec)
```

SUPPLIER

Supplier table has 4 rows: name, email, and lead time accompanied by supplier ID being the primary key.

```
mysql> describe supplier;
+----------------+--------------+------+-----+---------+----------------+
| Field          | Type         | Null | Key | Default | Extra          |
+----------------+--------------+------+-----+---------+----------------+
| Supplier_ID    | int          | NO   | PRI | NULL    | auto_increment |
| Company_Name   | varchar(100) | NO   |     | NULL    |                |
| Contact_Email  | varchar(100) | NO   |     | NULL    |                |
| Lead_Time_Days | int          | NO   |     | NULL    |                |
+----------------+--------------+------+-----+---------+----------------+
4 rows in set (0.00 sec)
```

SALE

Sale table has 6 rows: timestamp, total amount, and discount value accompanied by foreign keys: customer ID and employee ID followed by sale ID being the primary key.

```
mysql> describe sale;
+-----------------+---------------+------+-----+-------------------+-------------------+
| Field           | Type          | Null | Key | Default           | Extra             |
+-----------------+---------------+------+-----+-------------------+-------------------+
| Sale_ID         | int           | NO   | PRI | NULL              | auto_increment    |
| Timestamp       | datetime      | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| Employee_ID     | int           | NO   | MUL | NULL              |                   |
| Customer_ID     | int           | YES  | MUL | NULL              |                   |
| Total_Amount    | decimal(10,2) | NO   |     | NULL              |                   |
| Discount_Value  | decimal(10,2) | YES  |     | 0.00              |                   |
+-----------------+---------------+------+-----+-------------------+-------------------+
6 rows in set (0.00 sec)
```

PURCHASE ORDER

Purchase Order table has 5 rows: created at, status, and purchase order ID being the primary key accompanied by foreign keys: employee ID and supplier ID.

```
mysql> describe purchase_order;
+-------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| Field       | Type                                          | Null | Key | Default           | Extra             |
+-------------+-----------------------------------------------+------+-----+-------------------+-------------------+
| PO_ID       | int                                           | NO   | PRI | NULL              | auto_increment    |
| Created_At  | datetime                                      | YES  |     | CURRENT_TIMESTAMP | DEFAULT_GENERATED |
| Employee_ID | int                                           | NO   | MUL | NULL              |                   |
| Supplier_ID | int                                           | NO   | MUL | NULL              |                   |
| Status      | enum('Pending','Ordered','Received','Cancelled') | NO   |     | NULL              |                   |
+-------------+-----------------------------------------------+------+-----+-------------------+-------------------+
5 rows in set (0.00 sec)
```

SALE ITEM

Sale Item table has 4 rows: quantity, unit price with purchase order ID making up the composite key.

```
mysql> describe sale_item;
+-------------+---------------+------+-----+---------+-------+
| Field       | Type          | Null | Key | Default | Extra |
+-------------+---------------+------+-----+---------+-------+
| Sale_ID     | int           | NO   | PRI | NULL    |       |
| Product_ID  | int           | NO   | PRI | NULL    |       |
| Quantity    | int           | NO   |     | NULL    |       |
| Unit_Price  | decimal(10,2) | NO   |     | NULL    |       |
+-------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

PURCHASE ORDER ITEM

Purchase Order Item table has 4 rows: required quantity, unit price with purchase order ID and product ID together making up the composite key.

```
mysql> describe purchase_order_item;
+---------------+---------------+------+-----+---------+-------+
| Field         | Type          | Null | Key | Default | Extra |
+---------------+---------------+------+-----+---------+-------+
| PO_ID         | int           | NO   | PRI | NULL    |       |
| Product_ID    | int           | NO   | PRI | NULL    |       |
| Quantity_Req  | int           | NO   |     | NULL    |       |
| Unit_Cost     | decimal(10,2) | NO   |     | NULL    |       |
+---------------+---------------+------+-----+---------+-------+
4 rows in set (0.00 sec)
```

THANK YOU!